

WASP-35b

R-1596

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_171214 · created 2026-08-02T01:38:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458101.8373 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1127 +/- 0.0093
Transit depth (Rp/Rs)^2	1.27 +/- 0.21 [%]
Orbital Inclination (inc)	85.59 +/- 0.43
Airmass coefficient 1 (a1)	1.2211 +/- 0.0094
Airmass coefficient 2 (a2)	-0.1187 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	None
Optimal Aperture	4.07
Optimal Annulus	5.63
Transit Duration (day)	0.1097 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1127 $\pm$ 0.0093 vs 0.1241 (deviation 1.02 σ)
Duration fit vs published	0.1097 d vs 0.1317 d (deviation 3.14 σ)
Mid-time O–C	3.3 min
Depth SNR	10.1
Residual scatter	0.76 %

Light curve

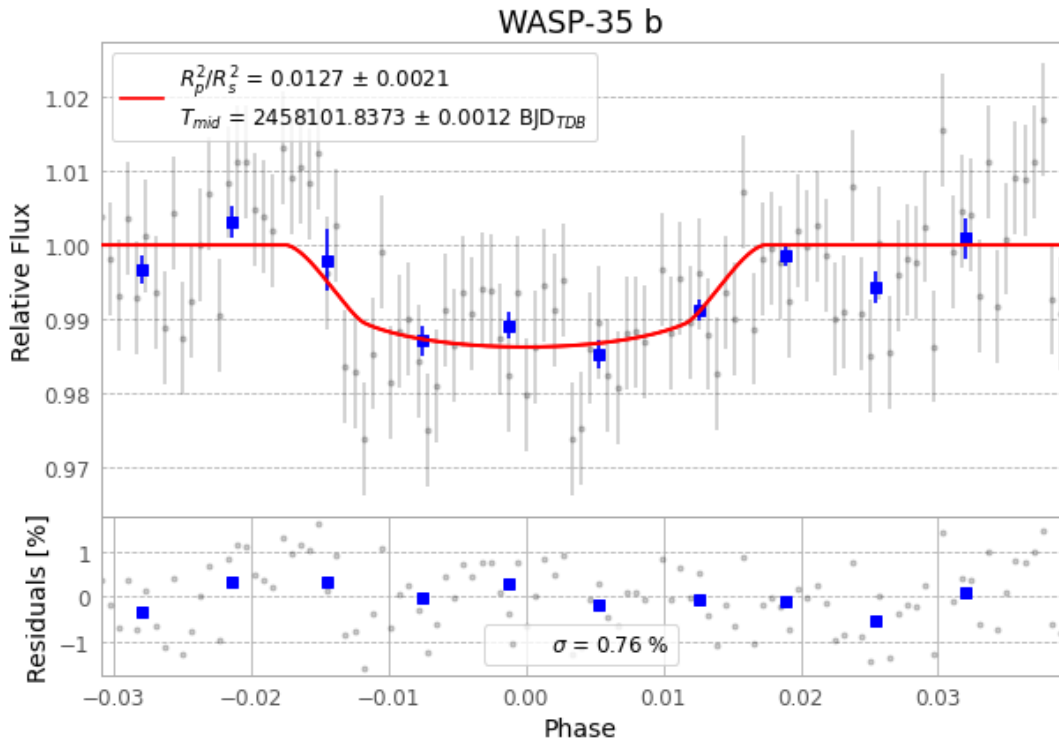


Figure 1 — FinalLightCurve_WASP-35 b_2017-12-13.png (SHA-256 c90bf5e92635f3bb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [218, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c177afe061afbcd7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-35171214053612.FITS → WASP-35171214105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-171214.log (14dc537d370f...), FinalLightCurve_WASP-35 b_2017-12-13.png (c90bf5e92635...), FinalLightCurve_WASP-35 b_2017-12-13.csv (4f594e722488...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:38Z.